

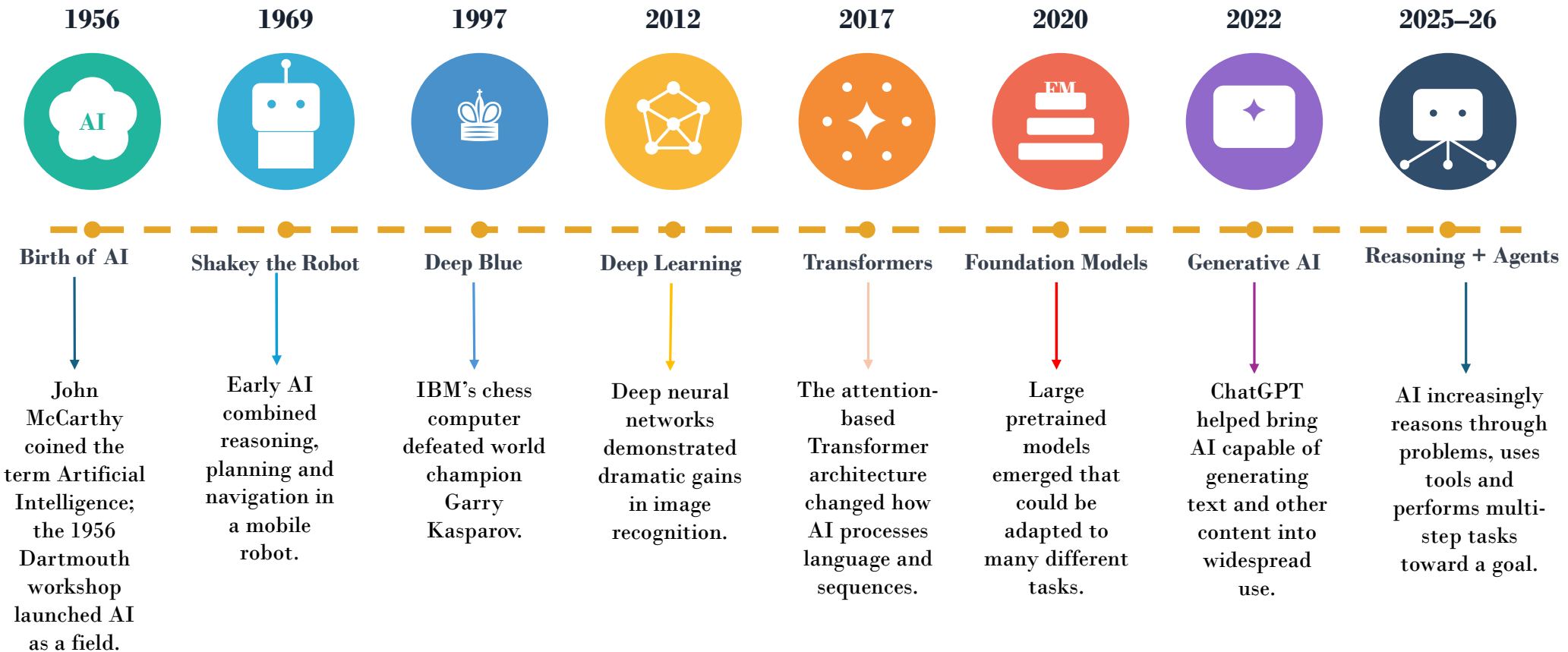
Week 1/Lesson 1: Foundations of AI & Machine Learning

- What is artificial intelligence?
- A Brief History and Evolution of AI
- AI vs. Traditional Programming and Automation
- AI, Machine Learning, and Deep Learning
- What is a machine learning model?
- Why Machine Learning?
- How Machine Learning Works
- Types of Machine Learning
- Scientific & Engineering Applications of AI/ML
- From Scientific Problem to ML Problem

What is artificial intelligence?

- Any technique that enables **machines** to mimic **human behavior** i.e., perform tasks that require human intelligence
- Tasks that normally require human intelligence:
 - ✓ speech recognition
 - ✓ object recognition
 - ✓ learning from data
 - ✓ problem solving
 - ✓ decision making

History and Evolution of Artificial Intelligence



RULES → PLANNING → SPECIALIZED AI → LEARNING FROM DATA → FOUNDATION MODELS → GENERATION → ACTION

Types of Artificial Intelligence

Purely Reactive



Reacts only to the current situation. No memory of past events.

Examples:

- Deep Blue
- Industrial Robots
- Rule-based spam filters

Limited Memory

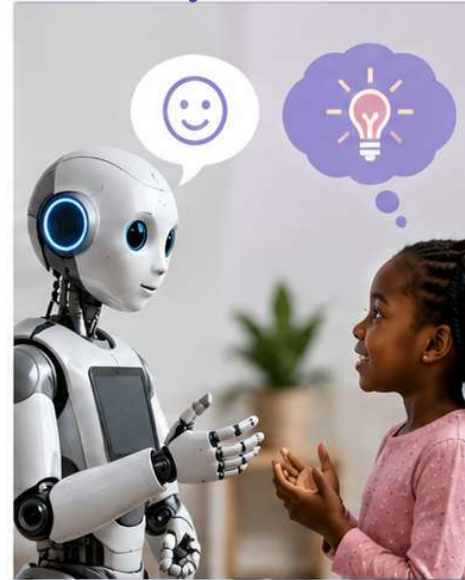


Uses recent data or past experience to inform current decisions.

Examples:

- Self-driving vehicles
- Recommender systems
- Predictive maintenance
- Medical diagnosis
- Conversational AI

Theory of Mind



Would understand beliefs, intentions and emotions that shape behaviour.

Examples:

- Social robots
- Modeling beliefs
- Infer human intentions

Research frontier

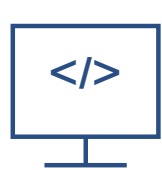
Self-Aware



Hypothetical AI with awareness of itself and its own mental states.

Hypothetical-does not currently exist

AI vs Automation vs Traditional Programming



TRADITIONAL PROGRAMMING

Humans write explicit instructions (rules, logic, algorithms). The computer follows the instructions step by step to produce the output.

INPUT

Data / Problem

+

PROGRAM

Human-written rules and instructions

=

OUTPUT

Result
(Deterministic)

EXAMPLES

- Payroll calculation
- Grade computation
- Inventory update
- Simple calculator

Follows
Explicit
rules



AUTOMATION

Uses technology to perform repetitive, rule-based tasks automatically. It executes predefined workflows with minimal human intervention.

INPUT

Trigger / Data

+

AUTOMATED WORKFLOW

Predefined steps and rules

=

OUTPUT

Completed Task / Action

EXAMPLES

- Email auto-responder
- Bank transaction processing
- Data entry automation
- Robotic process automation (RPA)

Executes
predefined
processes



ARTIFICIAL INTELLIGENCE (AI)

Systems that can learn from data, recognize patterns, make decisions, and improve over time—without being explicitly programmed for every situation.

INPUT

Data / Experience

+

LEARNING MODEL

Learns patterns from data and makes predictions or decisions

=

OUTPUT

Prediction / Decision / Insight

EXAMPLES

- Image recognition
- Disease prediction
- ChatGPT and language models
- Recommendation systems

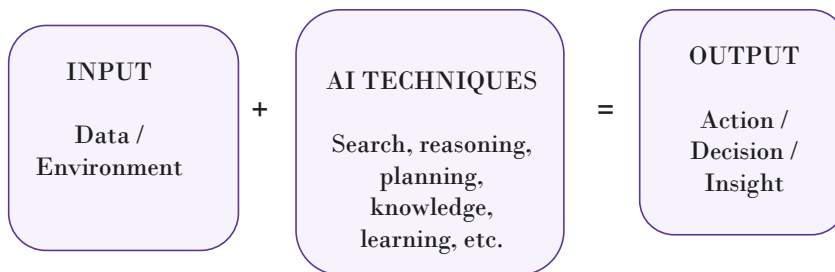
Learns
from data
and
adapts

Artificial Intelligence vs Machine Learning vs Deep Learning



ARTIFICIAL INTELLIGENCE (AI)

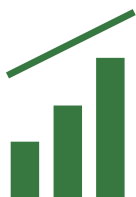
A broad field focused on creating systems that perform tasks normally requiring human intelligence.



EXAMPLES

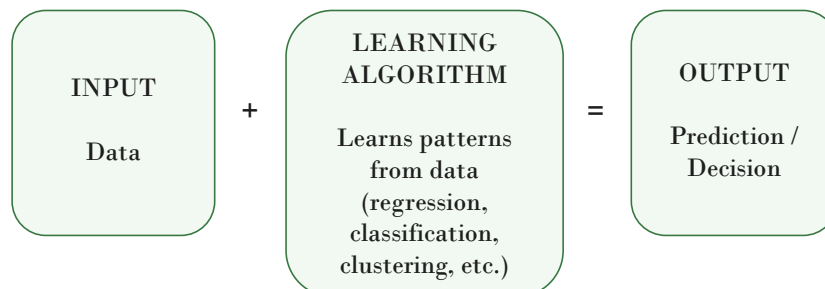
- Expert systems
- Robotics
- Game playing
- Natural language systems
- Planning & scheduling

Broad field:
creates
human-like
intelligent
behavior



MACHINE LEARNING (ML)

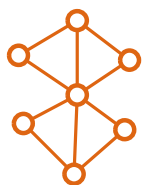
A subset of AI that enables computers to learn patterns from data and experience through algorithms to make predictions or decisions.



EXAMPLES

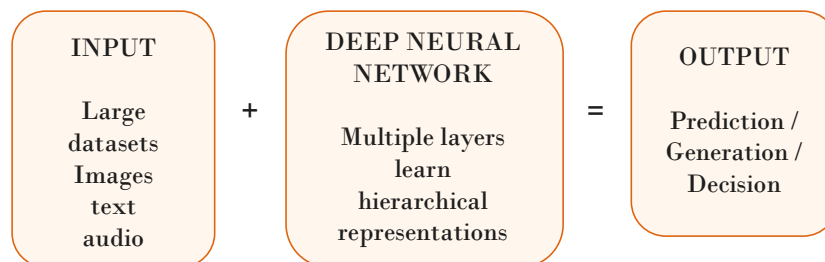
- Spam detection
- Fraud detection
- Recommendation systems
- Credit-risk scoring
- Predictive maintenance

Uses
algorithms to
learn from
data
to make
predictions
or decisions



DEEP LEARNING (DL)

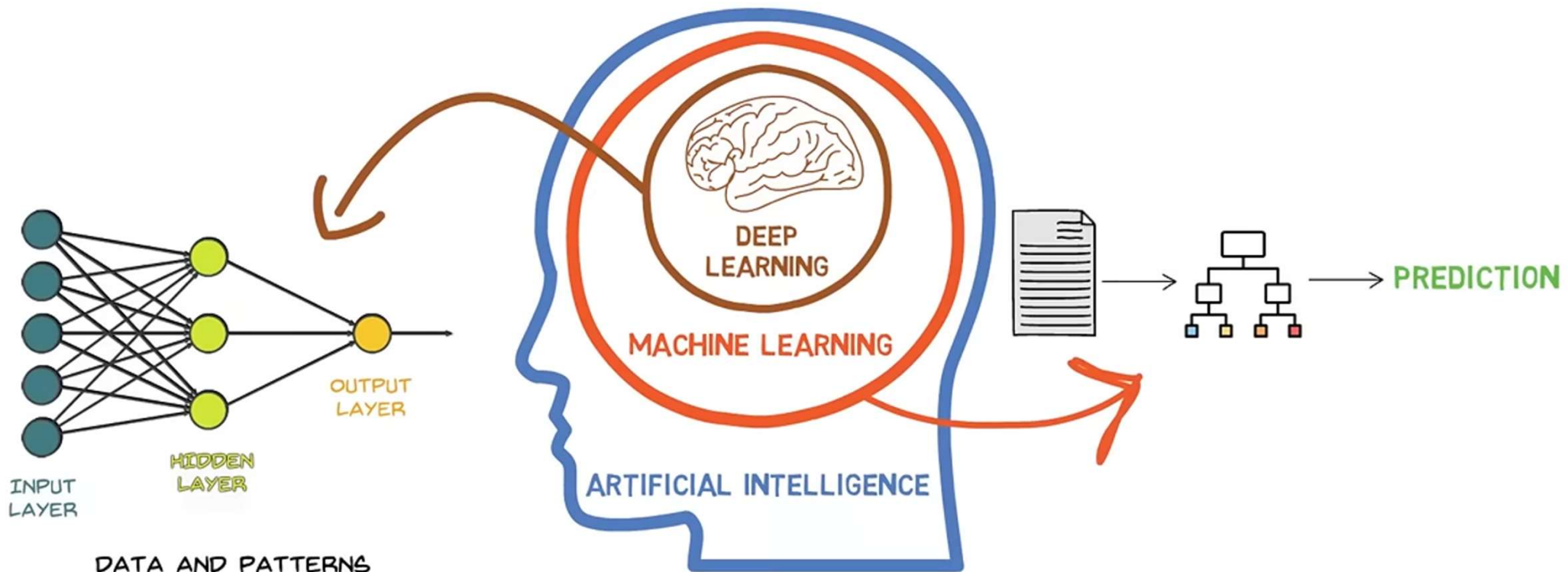
A subset of machine learning that uses multi-layer neural networks to learn complex patterns, especially from unstructured data.



EXAMPLES

- Image recognition
- Speech recognition
- Language generation
- Medical image analysis
- Autonomous perception

Uses deep
neural
networks to
learn
complex
patterns



DATA AND PATTERNS
CAN BE BETTER PERCEIVED

Machine Learning

Automatically extracting patterns from data to make predictions, decisions or discover structure

Algorithms learn relationships from examples rather than being given every rule explicitly.

MODELS

- ❖ Linear & logistic regression
- ❖ Decision trees & random forests
- ❖ Support vector machines
- ❖ k-nearest neighbours
- ❖ Bayesian models
- ❖ Neural networks (deep learning)

DATA / USE CASE TYPES

- ❖ Tabular / structured data
- ❖ Computer vision
- ❖ Natural language processing
- ❖ Time-series & forecasting
- ❖ Signal processing
- ❖ Robotics & control
- ❖ Scientific / spatial data

PARADIGMS

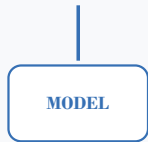
- ❖ Generative vs discriminative models
- ❖ Physics-informed & hybrid ML
- ❖ Interpretable / explainable ML
- ❖ Task-specific vs general-purpose models
- ❖ Probabilistic vs deterministic models

Types of Machine Learning

What does the training data or environment tell the learner?

SUPERVISED

LABELLED TRAINING DATA



New sample → CLAY

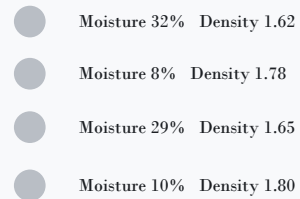
The model is shown the correct answer for each example.

Examples:

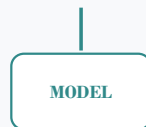
House-price prediction
Disease classification
Soil-strength prediction

UNSUPERVISED

UNLABELLED TRAINING DATA



No Clay/Sand labels



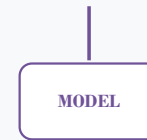
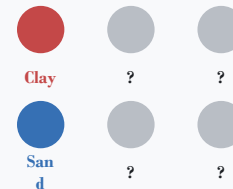
The model discovers groups without being told their names.

Examples:

Customer grouping
Grouping soil samples by properties
Anomaly detection

SEMI-SUPERVISED

A FEW LABELS



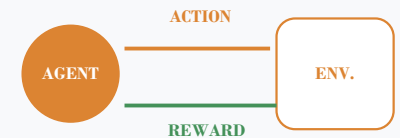
A small labelled set helps the model make use of much more unlabelled data.

Examples:

Medical-image classification with few labelled scans
Land-cover mapping with limited labels

REINFORCEMENT

INTERACTION + FEEDBACK



Good action +10

Poor action -5

The agent learns which actions lead to better rewards.

Examples:

Robot navigation
Game playing
Autonomous control

Supervised = labels

Unsupervised = no labels

Semi-supervised = some labels

Reinforcement = rewards

The Machine Learning Process

Development → Training → Trained Model → Inference

1 DEVELOPMENT

- Define the problem and target
- Prepare the data pipeline
- Choose evaluation metrics
- Compare algorithms & hyperparameters

2 TRAINING

- Feed representative training data
- Fit / adjust model parameters
- Evaluate and tune performance
- Retrain when data or objectives change

3 TRAINED MODEL

- Selected model
- Learned parameters
- Validated performance
- Ready for deployment

4 INFERENCE

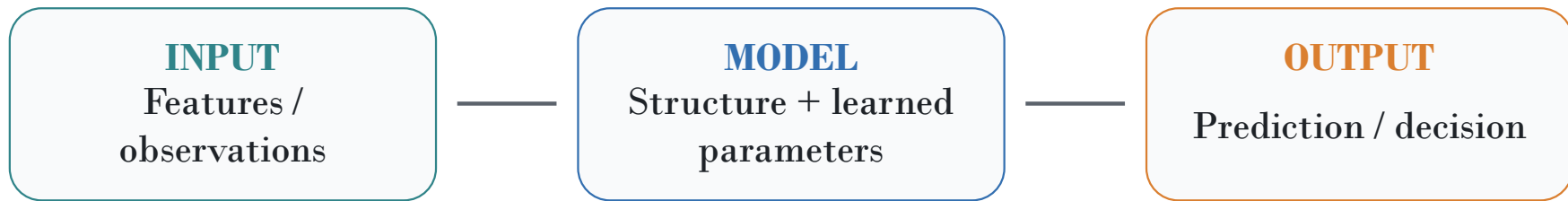
- Provide new / unseen inputs
- Apply the trained model
- Produce predictions or decisions
- Monitor performance and data drift

monitor → update → retrain when needed

What is a Machine Learning Model?

A model is a learned mapping from inputs to outputs...

A mathematical or computational relationship that maps inputs to outputs.



Simple linear regression model: $y = mx + c$

Multiple linear regression model: $y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$

Complex ML models: thousands/millions/billions of parameters

When Should You Use Machine Learning?

Machine learning is useful when learning from examples is more practical than explicitly writing the rules.



RULES ARE DIFFICULT TO WRITE

The relationship is too complex, variable, or poorly understood to encode reliably as explicit rules.



RELEVANT DATA ARE AVAILABLE

You have enough representative, good-quality data that reflect the problem and conditions in which the model will be used.



UNCERTAINTY IS ACCEPTABLE

Predictions or estimates are useful even when they are not perfect, and errors can be measured, evaluated, and managed.



VALUE JUSTIFIES THE COST

The expected benefit outweighs the cost of collecting data, developing the model, deploying it, and maintaining it.

NOT EVERY PROBLEM NEEDS ML • If explicit rules solve it reliably and efficiently, use them.

Strengths & Limitations of Machine Learning

STRENGTHS

- ✓ **Finds patterns**
Detects relationships that may be difficult to specify manually.
- ✓ **Scales well**
Can process large, complex and heterogeneous datasets.
- ✓ **Predicts & optimises**
Useful for forecasting, classification, and complex system optimization.
- ✓ **Works continuously**
Once deployed, models can make predictions repeatedly and quickly.
- ✓ **Integrates methods**
Can complement physics-based, statistical and domain models.

LIMITATIONS

- ! **Not for every problem**
A simpler rule, equation or conventional program may be better.
- ! **Can be wrong**
High confidence does not guarantee a correct prediction.
- ! **Needs good data**
Poor, biased or unrepresentative data produce poor models.
- ! **Can be expensive**
Some models require substantial data, computation and energy.
- ! **Patterns can change**
Performance may deteriorate when real-world conditions shift.

Applications of Artificial Intelligence by Sector

Healthcare Disease diagnosis • Medical imaging • Drug discovery	Agriculture Crop monitoring • Yield prediction • Pest detection	Engineering & Construction Structural health monitoring • Predictive maintenance • Design optimization
Academia & Research Data analysis • Literature review • Scientific discovery	Business & Finance Fraud detection • Risk assessment • Customer insights	Education Personalized learning • AI tutoring • Automated assessment
Environment & Climate Climate modeling • Disaster prediction • Environmental monitoring	Transportation Autonomous vehicles • Traffic prediction • Route optimization	Government & Public Services Policy analysis • Public service delivery • Smart cities
Manufacturing Quality control • Robotics • Predictive maintenance	Media & Creative Industries Content generation • Image/video creation • Recommendation systems	Everyday Life Voice assistants • Search • Translation • Smart homes

Common Use Cases of AI/ML

Across different sectors, AI/ML is often used to perform similar kinds of tasks.

1 Computer Vision & Image Analysis

Extract information from images and video.

2 Natural Language Processing (NLP)

Understand, extract and generate information from language.

3 Prediction & Forecasting

Estimate unknown or future outcomes from data.

4 Pattern Discovery & Anomaly Detection

Find hidden structure, relationships and unusual behaviour.

5 Optimization & Decision-Making

Identify better actions, designs or operating conditions.

6 Scientific Modelling, Discovery & Experimentation

Accelerate modelling, exploration, discovery and experiments.

These use cases overlap — a single AI/ML system may serve more than one purpose.

a) Computer Vision & Image Analysis

How can AI/ML extract useful information from images and video?

HOW IT WORKS



EXAMPLES

Infrastructure

Detect cracks and defects in bridges, roads and buildings.

Agriculture

Identify crop disease from leaf images or drone imagery.

Healthcare

Detect or segment abnormalities in medical scans.

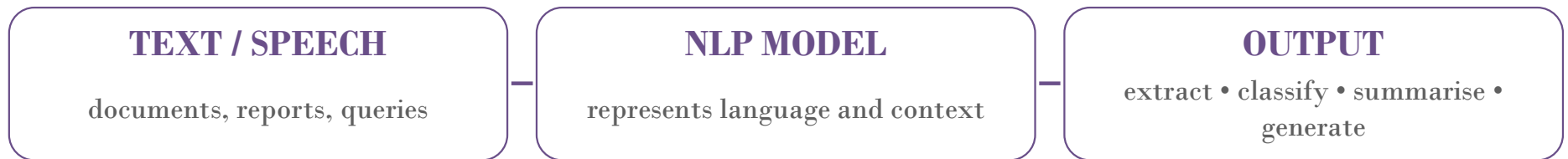
Environment

Map floods, land cover or environmental change from satellite images.

b) Natural Language Processing (NLP)

How can AI/ML work with scientific, technical and everyday language?

HOW IT WORKS



EXAMPLES

Scientific literature

Extract materials, properties and relationships from thousands of papers.

Engineering records

Classify incident reports and extract recurring causes or hazards.

Healthcare

Extract information from clinical notes and medical records.

Knowledge assistants

Search, summarise and answer questions over technical documents.

c) Prediction & Forecasting

How can AI/ML estimate an unknown value or a future outcome?

HOW IT WORKS



EXAMPLES

Geotechnical engineering

Predict soil shear strength from soil properties and test data.

Environment

Forecast rainfall, river level, flooding or other hazards.

Manufacturing

Predict equipment failure or remaining useful life from sensor data.

Energy & agriculture

Forecast electricity demand, crop yield or other time-dependent outcomes.

d) Pattern Discovery & Anomaly Detection

How can AI/ML reveal structure or unusual behaviour in complex data?

HOW IT WORKS



EXAMPLES

Earth materials

Group soil or rock samples with similar measured properties.

Machine monitoring

Detect unusual vibration patterns before equipment failure.

Environmental sensing

Flag abnormal water-quality or environmental sensor readings.

Research data

Reveal patterns in large multidimensional experimental datasets.

e) Optimization & Decision-Making

How can AI/ML help choose better actions, designs or operating conditions?

HOW IT WORKS



EXAMPLES

Materials & construction

Optimize concrete mixes for strength, cost and embodied carbon.

Agriculture

Optimize irrigation or input schedules using soil and weather information.

Engineering design

Search design alternatives for improved performance under constraints.

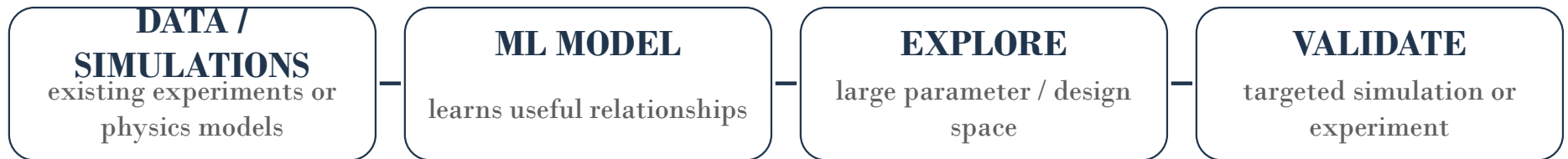
Control & autonomy

Support robot control, route planning and process operation.

f) Scientific Modelling, Discovery & Experimentation

How can AI/ML accelerate modelling, explore possibilities and guide experiments?

HOW IT WORKS



EXAMPLES

Surrogate modelling

Approximate expensive FEM, DEM or CFD simulations for rapid evaluation.

Parametric studies

Explore many parameter combinations without running every costly simulation.

Materials discovery

Screen large candidate spaces and identify promising materials for testing.

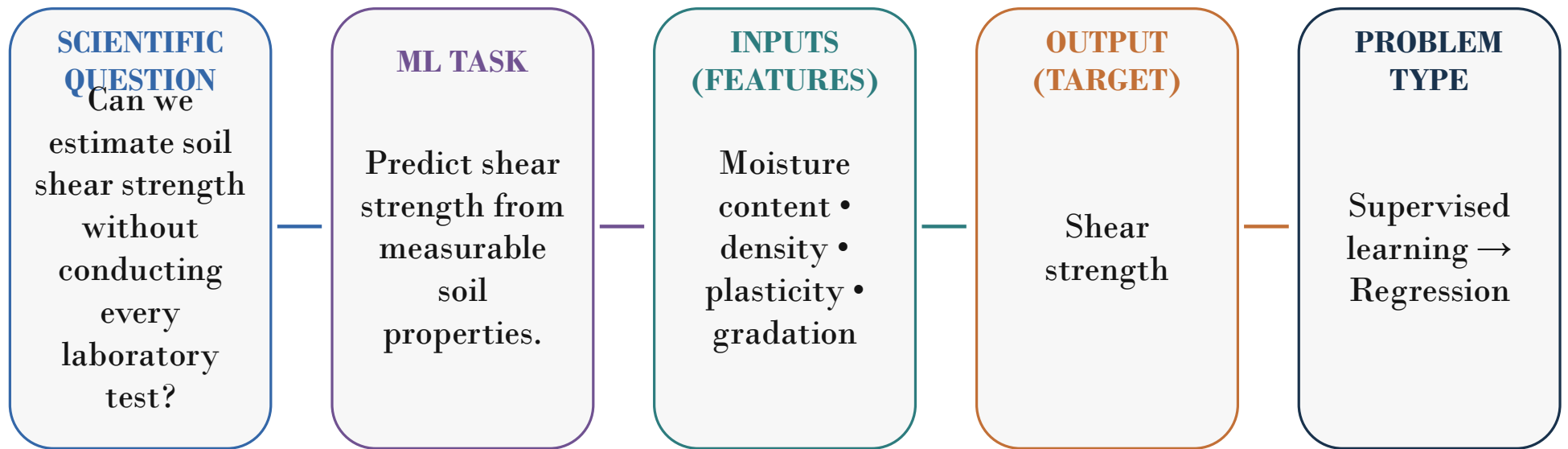
ML-guided experiments

Prioritize informative or promising experiments, then use new results to refine the model.

ML can reduce the search space and experimental burden; physical validation remains essential.

From Scientific Problem to ML Problem

Before choosing an algorithm, translate the scientific question into a problem that data and ML can address.



Key question: What are the inputs, what is the desired output, and what data can we obtain?

Problem formulation comes before model selection.

Week 1 Assignment | Problem Identification

Complete the Week 1 project task using the attached Google Form:

<https://docs.google.com/forms/d/e/1FAIpQLSfk0rZB39d04KA1Q8kzdGitNcQ27UPBgqSbG05coZaA9xN-zA/viewform?usp=preview&authuser=0>

Assignment due: 22nd September 2026